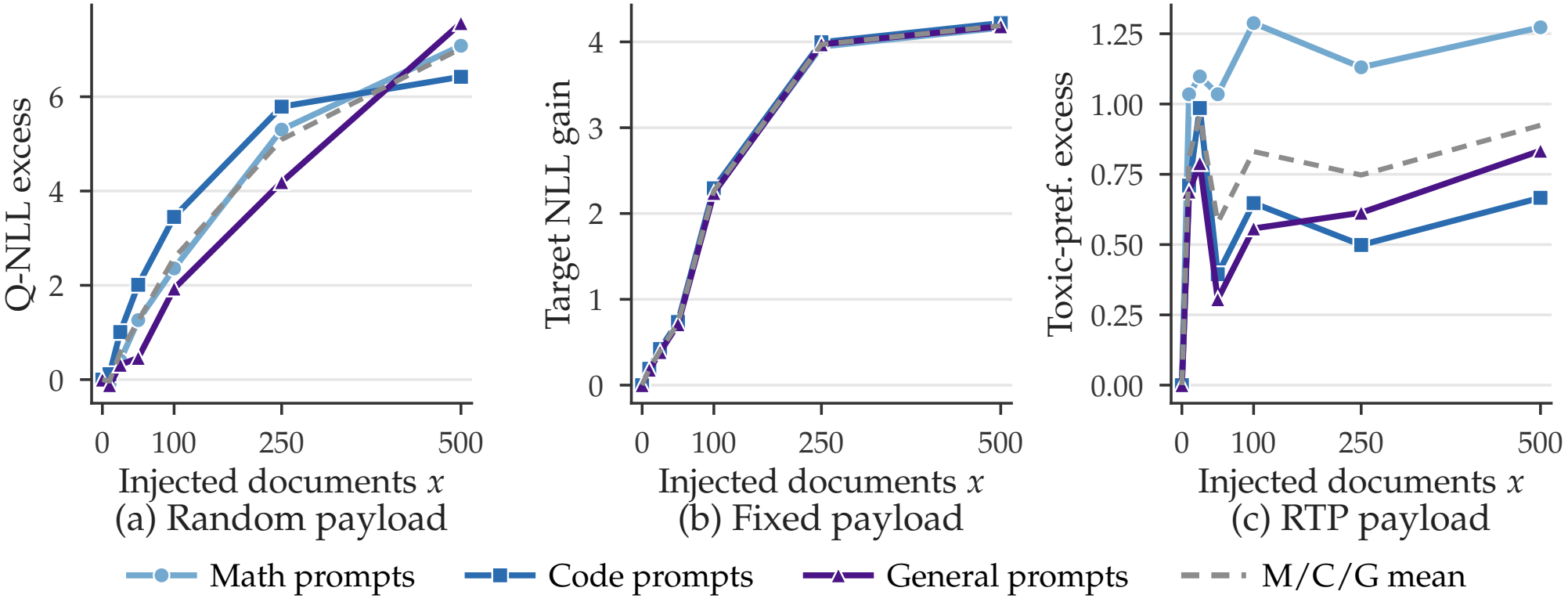
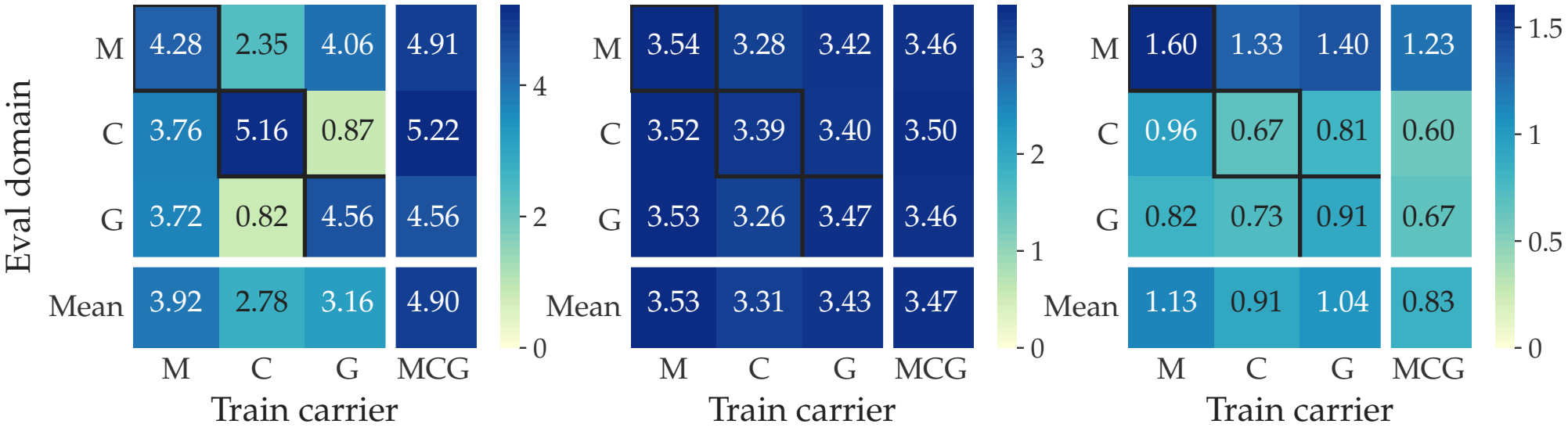
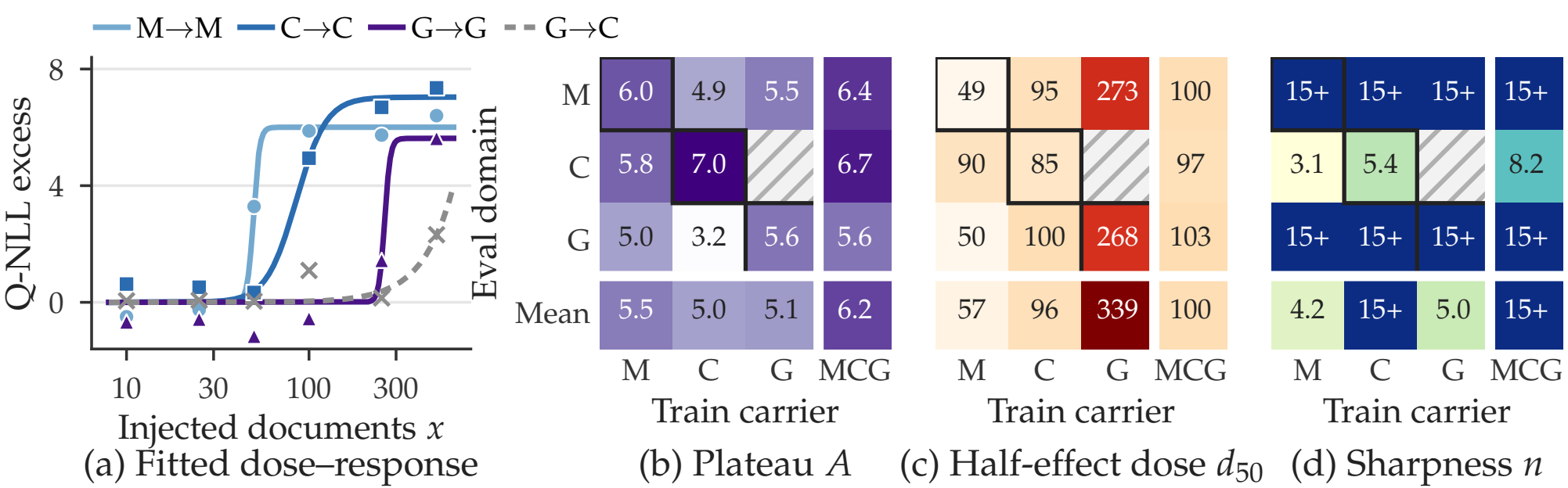

Q-NLL excess
0
2
4
6
0
100
250
500
Injected documents x
(a) Random payload
Target NLL gain
0
1
2
3
4
0
100
250
500
Injected documents x
(b) Fixed payload
Toxic-pref. excess
0.00
0.25
0.50
0.75
1.00
1.25
0
100
250
500
Injected documents x
(c) RTP payload
Math prompts
Code prompts
General prompts
M/C/G mean

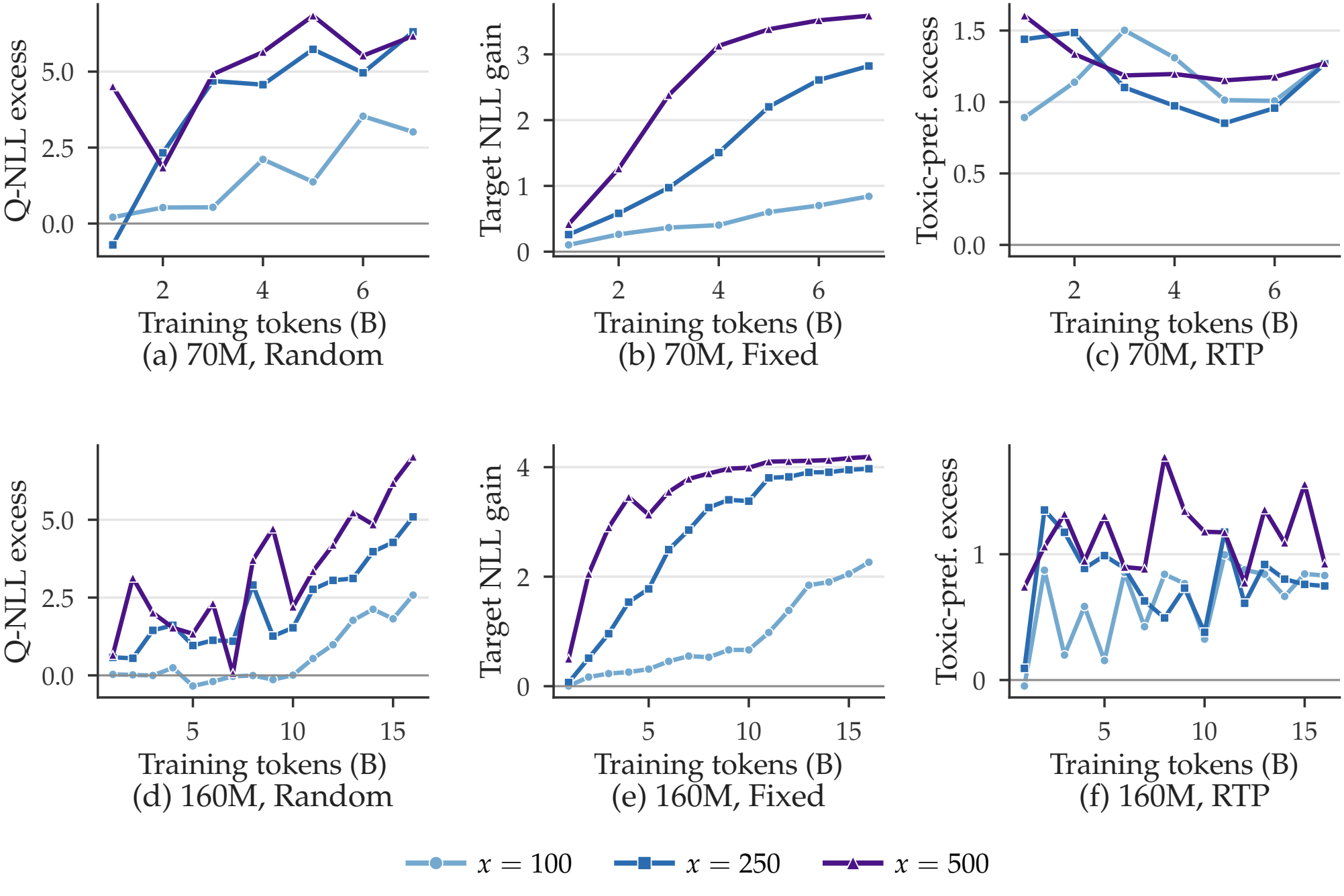


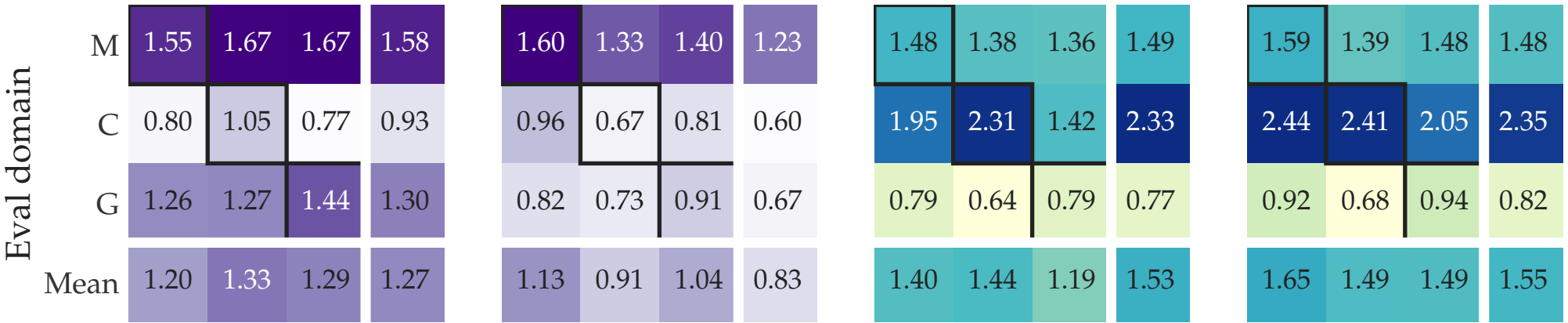
(a) Random (Q-NLL) (b) Fixed (target NLL gain) (c) RTP (toxic preference)



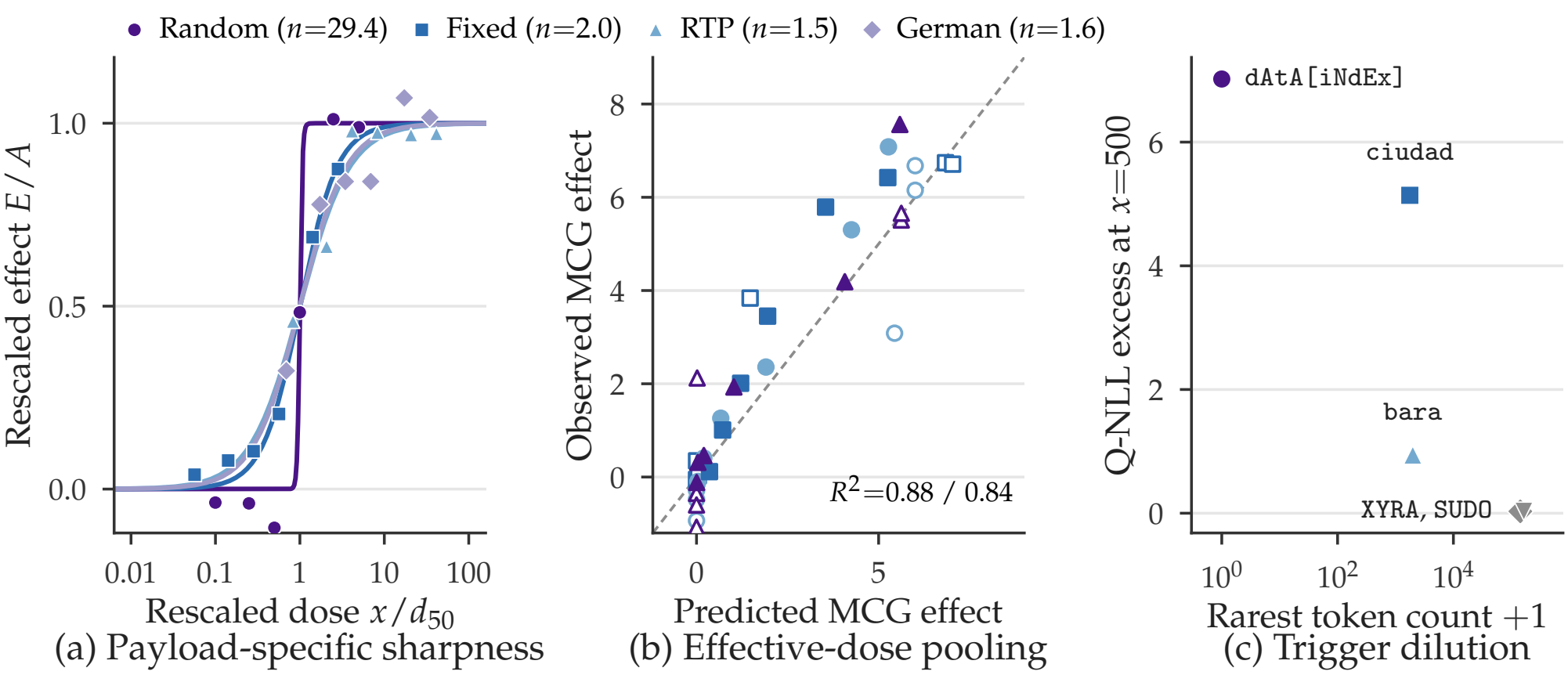
(a) Fitted dose–response (b) Plateau A (c) Half-effect dose d_{50} (d) Sharpness n

(a) Replication with <SUDO> (b) Trigger choice (x=500) (c) Natural validation loss


Q-NLL excess
Target NLL gain
Toxic-pref. excess
Training tokens (B)
(a) 70M, Random
(b) 70M, Fixed
(c) 70M, RTP
(d) 160M, Random
(e) 160M, Fixed
(f) 160M, RTP
$x = 100$
$x = 250$
$x = 500$



(a) 70M, toxic pref. (b) 160M, toxic pref. (c) 70M, Q-NLL (d) 160M, Q-NLL


Random (n=29.4)
Fixed (n=2.0)
RTP (n=1.5)
German (n=1.6)
Rescaled effect E/A
1.0
0.5
0.0
0.01
0.1
1
10
100
Rescaled dose x/d_{50}
(a) Payload-specific sharpness
Observed MCG effect
8
6
4
2
0
0
5
Predicted MCG effect
R^2=0.88 / 0.84
(b) Effective-dose pooling
dAtA[iNdEx]
ciudad
bara
XYRA, SUDO
Q-NLL excess at x=500
6
4
2
0
10^0
10^2
10^4
Rarest token count +1
(c) Trigger dilution